1. Introduction

This project simulates a small-scale enterprise network environment to demonstrate core cybersecurity operations, including system deployment, attack simulation, defensive configuration, and monitoring.

A vulnerable web application (DVWA) and a honeypot were deployed to emulate real-world systems. Various attacks such as network scanning and web exploitation were conducted using industry-standard tools. Security controls including a Web Application Firewall (WAF) were implemented to mitigate these attacks, while monitoring tools were used to detect and analyze malicious activities.

The outcome highlights how vulnerabilities can be exploited, detected, and mitigated within a controlled environment.

2. Environment Setup

Network Architecture

- Attacker Machine: Kali Linux
- Target Server: Ubuntu Server hosting DVWA
- Monitoring Server: ELK Stack
- Honeypot System: Cowrie (SSH Honeypot)
- **NETWORK DIAGRAM HERE]**

```
┌──────────────────┐
│   Attacker    │
│  (Kali Linux) │
└──────────┬───────┘
        │  (Scan / Exploit)
        ▼
┌──────────────────────────┐
│   Target Server       │
│   (Ubuntu + DVWA)     │
│                       │
│   Web App (DVWA)      │
```

```
│ WAF (ModSecurity) │
│ Honeypot (Cowrie) │
└──────────┬──────────┘
           │ (Logs)
           ▼
┌──────────────────┐
│ Monitoring       │
│ (ELK Stack)      │
└──────────────────┘
```

3.Objectives

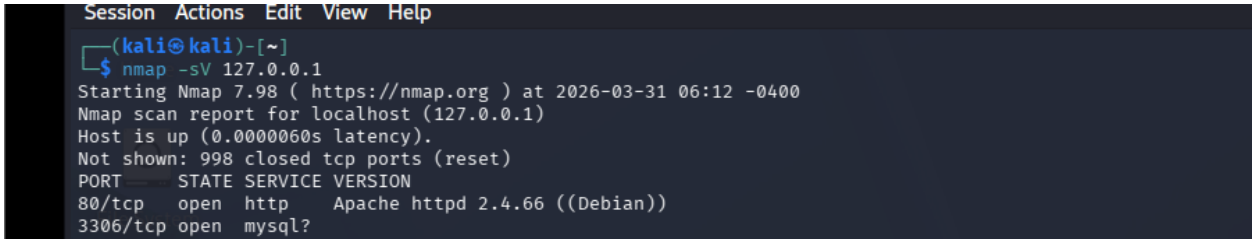
- Deploy a vulnerable web application (DVWA)
- Set up a honeypot to capture attacker behavior
- Simulate attacks using Nmap and Metasploit
- Implement defensive mechanisms (WAF/Firewall)
- Monitor system activities using logging tools (ELK Stack)
- Produce an incident response report based on findings

3.0 Attack Simulation

Network Scanning (Nmap)

- Attackers use Nmap to scan for open ports and services.
- Screenshot evidence: shows ports like 22 (SSH) and 80 (HTTP) open.
- Meaning: These are entry points attackers may try to exploit.

Config 1:1 Nmap scan and the open ports



```
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ nmap -sV 127.0.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-31 06:12 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000060s latency).
Not shown: 998 closed tcp ports (reset)
PORT     STATE SERVICE VERSION
80/tcp   open  http    Apache httpd 2.4.66 ((Debian))
3306/tcp open  mysql?
```

3.1 Metasploit Exploitation Attempt

- Attackers use Metasploit to exploit DVWA's command injection vulnerability.
- Screenshot evidence: shows attempts to run malicious commands.
- Some attempts succeed, others are blocked.
- Meaning: Demonstrates how attackers try to break in.

Config 1:2 exploit on DVWA's command injection vulnerability.

```
msf > use exploit/multi/http/apache_normalize_path_rce
[*] Using configured payload linux/x64/meterpreter/reverse_tcp
msf exploit(multi/http/apache_normalize_path_rce) >
```

input to this VM, move the mouse pointer inside or press Ctrl+G.

```
kali@kali: ~
Session  Actions  Edit  View  Help

Interact with a module by name or index. For example info 72, use 72 or use exploit/multi/http/zpanel_information_disclosure_rce
After interacting with a module you can manually set a TARGET with set TARGET 'Linux x86'

msf > search apache

Matching Modules
================

   #    Name                                                       Disclosure Date  Rank       Check  Description
   -    ----                                                       ---------------  ----       -----  -----------
   0    exploit/multi/http/apache_apisix_api_default_token_rce     2020-12-07       excellent  Yes    APISIX Admin API default acces
s token RCE
   1    exploit/linux/http/atutor_filemanager_traversal            2016-03-01       excellent  Yes    ATutor 2.2.1 Directory Travers
al / Remote Code Execution
   2    exploit/multi/http/apache_activemq_upload_jsp              2016-06-01       excellent  No     ActiveMQ web shell upload
   3      \_ target: Java Universal                                .                .          .      .
   4      \_ target: Linux                                         .                .          .      .
   5      \_ target: Windows                                       .                .          .      .
   6    auxiliary/scanner/http/apache_userdir_enum                 .                normal     No     Apache "mod_userdir" User Enum
eration
   7    exploit/multi/http/apache_normalize_path_rce               2021-05-10       excellent  Yes    Apache 2.4.49/2.4.50 Traversal
 RCE
   8      \_ target: Automatic (Dropper)                           .                .          .      .
   9      \_ target: Unix Command (In-Memory)                      .                .          .      .
   10   auxiliary/scanner/http/apache_normalize_path               2021-05-10       normal     No     Apache 2.4.49/2.4.50 Traversal
 RCE scanner
   11     \_ action: CHECK_RCE                                     .                .          .      Check for RCE (if mod_cgi is e
nabled).
   12     \_ action: CHECK_TRAVERSAL                               .                .          .      Check for vulnerability.
   13     \_ action: READ_FILE                                     .                .          .      Read file on the remote server
.
   14   exploit/windows/http/apache_activemq_traversal_upload      2015-08-19       excellent  Yes    Apache ActiveMQ 5.x-5.11.1 Dir
ectory Traversal Shell Upload
```

To direct input to this VM, click inside or press Ctrl+G.

```
   17        \_ target: Windows In-Memory                                                  .          .         .      .
   18     exploit/multi/http/avideo_notify_ffmpeg_unauth_rce                              2025-12-19 excellent Yes    AVideo n
otify.ffmpeg.json.php Unauthenticated RCE via Salt Discovery
   19        \_ target: Automatic                                                          .          .         .      .
   20        \_ target: PHP In-Memory                                                      .          .         .      .
   21        \_ target: Unix/Linux Command Shell                                           .          .         .      .
   22        \_ target: Windows Command Shell                                              .          .         .      .
   23     exploit/unix/webapp/awstatstotals_multisort                                      2008-08-26 excellent Yes    AWStats
Totals multisort Remote Command Execution
   24     exploit/unix/webapp/awstats_migrate_exec                                         2006-05-04 excellent Yes    AWStats
migrate Remote Command Execution
   25     exploit/multi/http/activecollab_chat                                             2012-05-30 excellent Yes    Active C
ollab "chat module" Remote PHP Code Injection Exploit
   26     exploit/unix/webapp/actualanalyzer_ant_cookie_exec                               2014-08-28 excellent Yes    ActualAn
alyzer 'ant' Cookie Command Execution
   27     exploit/windows/browser/adobe_flash_otf_font                                     2012-08-09 normal    No     Adobe Fl
ash Player 11.3 Kern Table Parsing Integer Overflow
   28        \_ target: Automatic                                                          .          .         .      .
   29        \_ target: IE 6 on Windows XP SP3                                             .          .         .      .
   30        \_ target: IE 7 on Windows XP SP3                                             .          .         .      .
   31        \_ target: IE 8 on Windows XP SP3                                             .          .         .      .
   32        \_ target: IE 7 on Windows Vista SP2                                          .          .         .      .
   33        \_ target: IE 8 on Windows 7 SP1                                              .          .         .      .
   34        \_ target: IE 9 on Windows 7 SP1                                              .          .         .      .
   35     exploit/windows/browser/adobe_flashplayer_arrayindexing                          2012-06-21 great     No     Adobe Fl
ash Player AVM Verification Logic Array Indexing Code Execution
   36     exploit/windows/browser/adobe_flash_regex_value                                  2013-02-08 normal    No     Adobe Fl
ash Player Regular Expression Heap Overflow
   37     exploit/unix/webapp/aerohive_netconfig_lfi_log_poison_rce                        2020-02-17 excellent Yes    Aerohive
 NetConfig 10.0r8a LFI and log poisoning to RCE
   38        \_ target: Linux                                                              .          .         .      .
   39        \_ target: CMD                                                                .          .         .      .
   40     exploit/multi/http/agent_tesla_panel_rce                                         2019-08-14 excellent Yes    Agent Te
sla Panel Remote Code Execution
   41     exploit/windows/misc/agentxpp_receive_agentx                                     2010-04-16 good      No     AgentX++
 Master AgentX::receive_agentx Stack Buffer Overflow
   42     exploit/multi/http/ajaxplorer_checkinstall_exec                                  2010-04-04 excellent Yes    AjaXplor
er checkInstall.php Remote Command Execution
   43     auxiliary/gather/alienvault_iso27001_sqli                                        2014-03-30 normal    No     AlienVau
lt Authenticated SQL Injection Arbitrary File Read
```

3.2 Web Application Exploitation (DVWA)

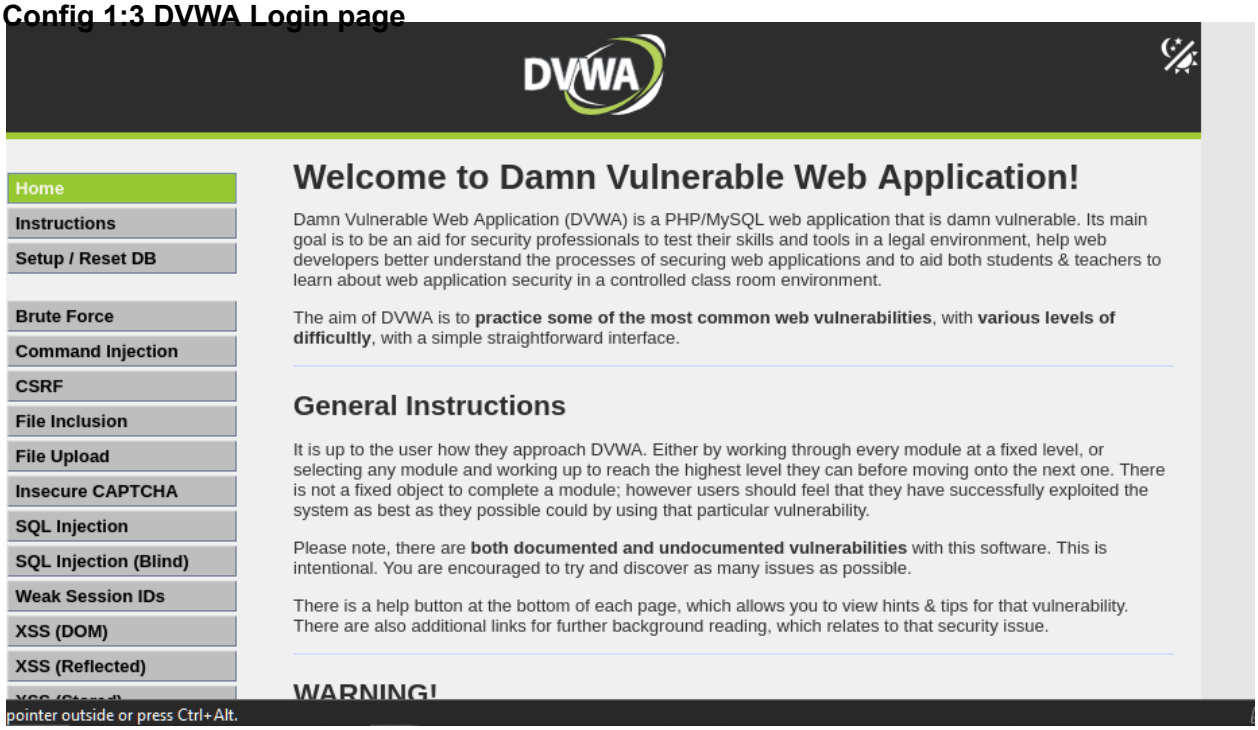
DVWA was installed on an Ubuntu Apache server with MySQL database configuration. The application was configured and tested successfully via a web browser.

SQL Injection

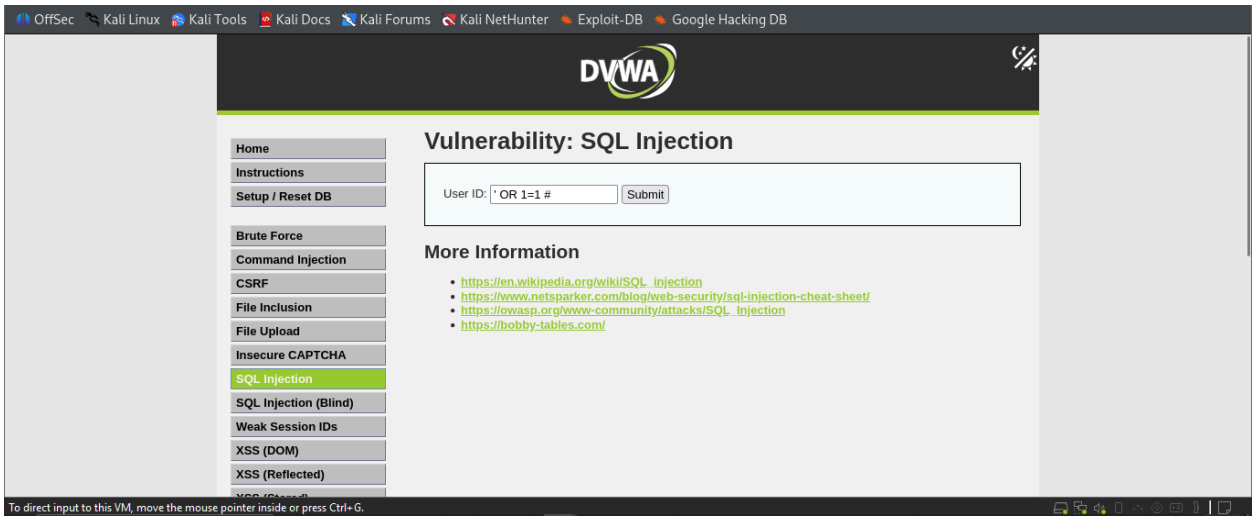
The login form was bypassed successfully using SQL Injection payload:

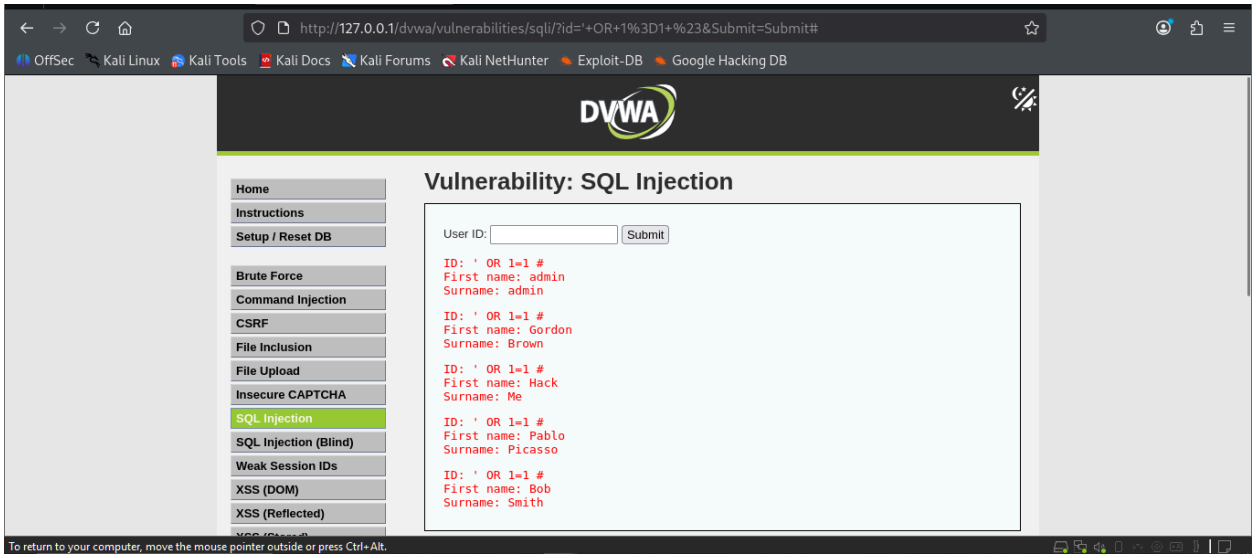
' OR '1'='1

Config 1:3 DVWA Login page



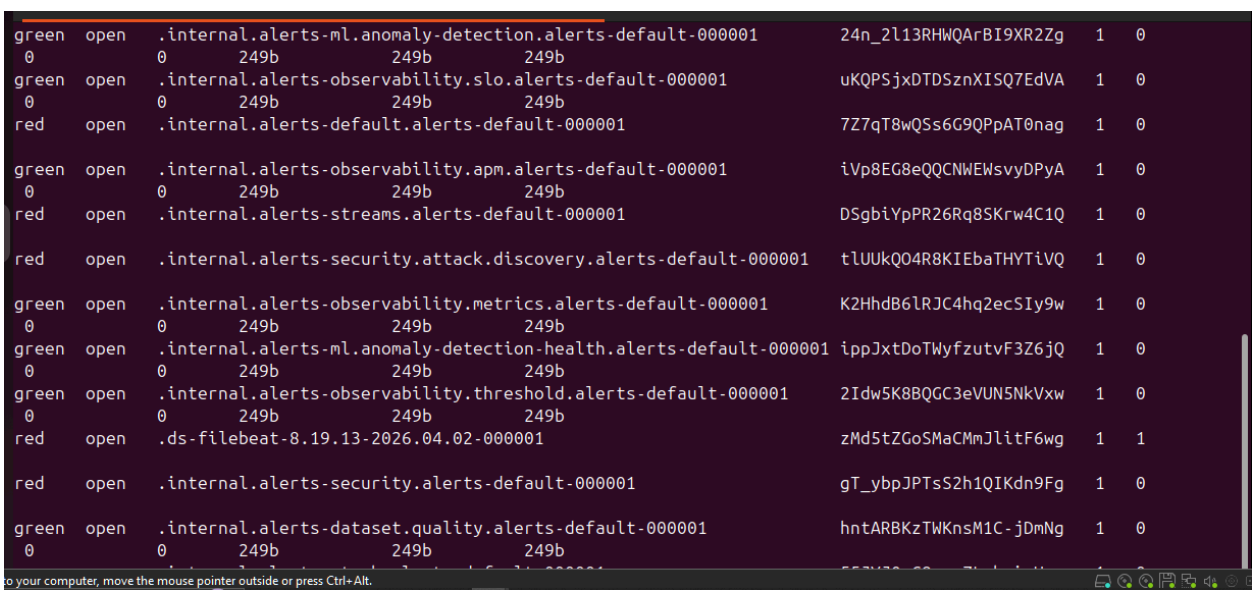
Config 1:4 successfully SQL Injection payload:





3.3 Honeypot Deployment

A Cowrie SSH honeypot was deployed to capture unauthorized login attempts and attacker interaction



Config 1:5 Honeypot Running / Logs]

3.4 ModSecurity Blocking Logs

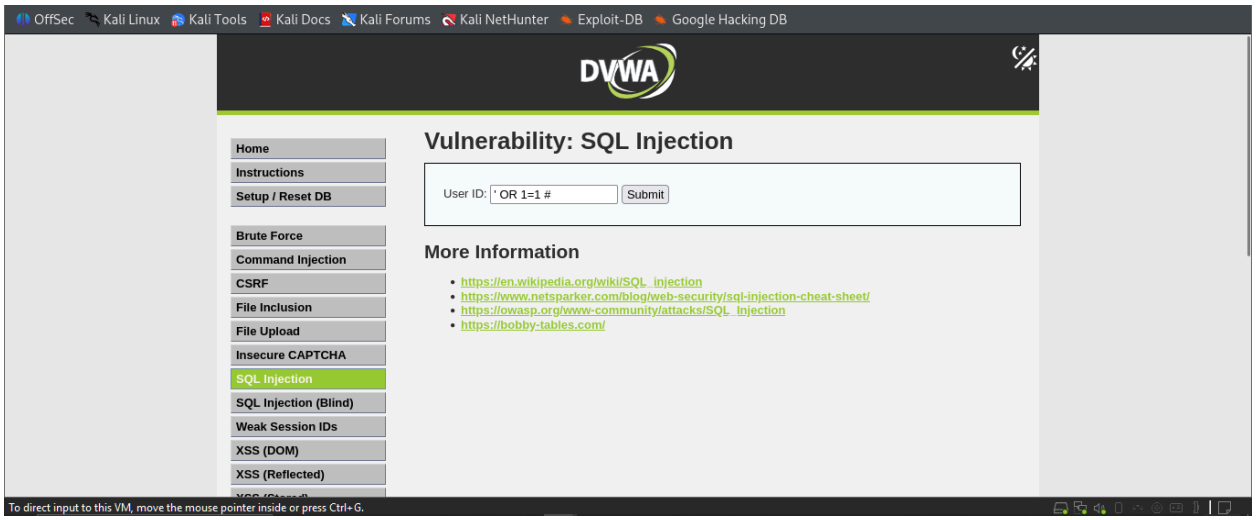
- ❖ ModSecurity detects and blocks suspicious requests.
- ❖ Screenshot evidence: shows blocked requests from ModSecurity
- ❖ Meaning: Confirms the firewall is working and protecting DVWA.

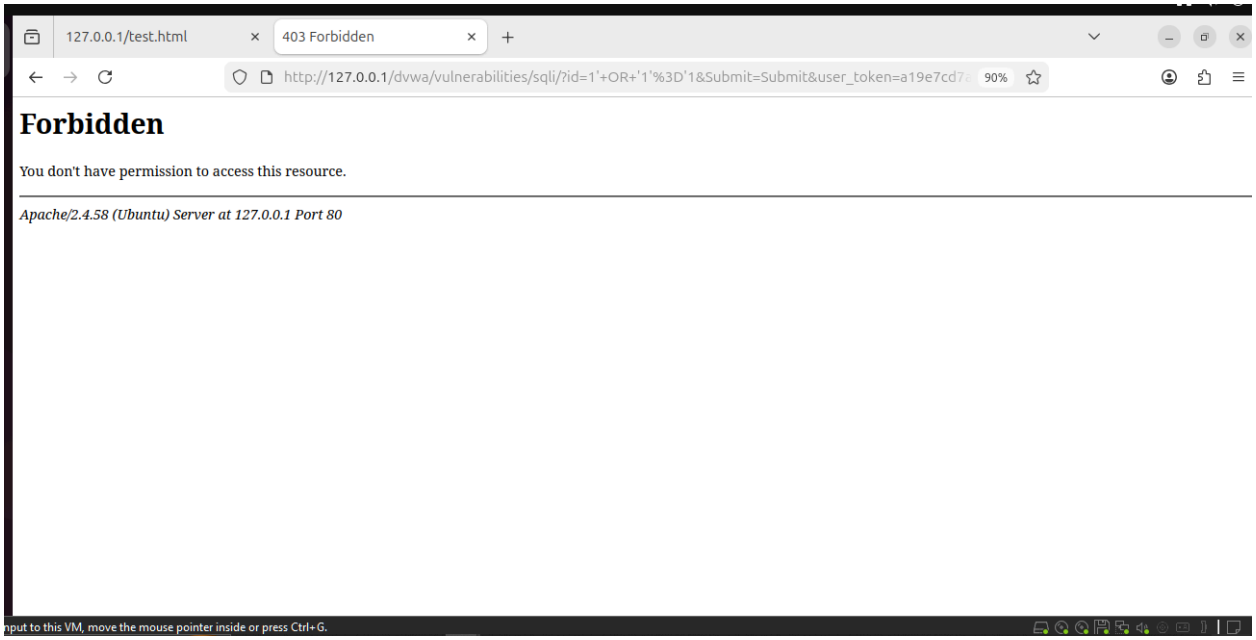
Config 1:5 shows Modsecurity setting

```
GNU nano 7.2                                    /etc/modsecurity/modsecurity.conf
# -- Rule engine initialization ----------------------------------------------

# Enable ModSecurity, attaching it to every transaction. Use detection
# only to start with, because that minimises the chances of post-installation
# disruption.
#
SecRuleEngine On
```

Config 1:6 shows Modsecurity blocking suspicious requests

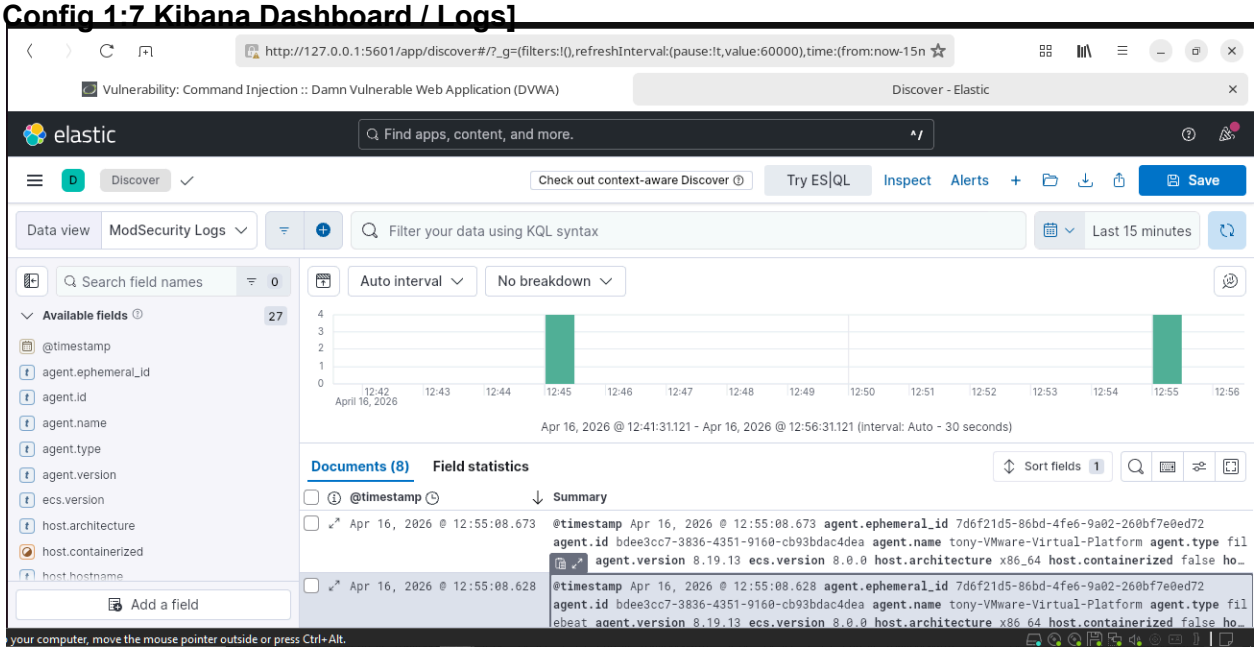




3.3 Monitoring Setup

System and application logs were configured for monitoring using ELK Stack (Elasticsearch, Logstash, Kibana).

Config 1:7 Kibana Dashboard / Logs]



3.4 Conclusion

This project successfully demonstrated the lifecycle of a cyber attack, from reconnaissance to exploitation and mitigation. It highlights the importance of proactive security measures, real-time monitoring, and proper incident response strategies in protecting enterprise environments.

3.5 Recommendations

- Enable more Filebeat modules to capture richer fields (like IP addresses and status codes).
- Add more charts (top URLs attacked, user agents).
- Set alerts in Kibana to notify admins when suspicious spikes occur.
- Integrate honeypot logs directly into Elasticsearch for unified monitoring.